

Sample Usage Guide for Agentic RAG Assistant

This guide provides practical examples of how to use the Agentic RAG Assistant effectively.

Getting Started - Step by Step

Step 1: Launch the Application

```
cd agentic_rag_app
streamlit run app.py
```

Navigate to `http://localhost:8501` in your browser.

Step 2: Upload Your First Documents

Example Document Types:

1. **Business Reports** (PDF)
2. **Meeting Minutes** (DOCX)
3. **Financial Data** (XLSX)
4. **Project Plans** (PPTX)
5. **Research Papers** (PDF)

Upload Process:

1. Go to " Document Upload" tab
2. Drag and drop or browse for files
3. Click " Process Documents"
4. Wait for processing to complete

Step 3: Ask Your First Questions

Start with simple questions to understand the system:

- "What documents have been uploaded?"
- "Give me a summary of the main topics."

Example Use Cases

1. Business Intelligence Analysis

Documents to Upload:

- Quarterly reports (PDF)
- Sales data (XLSX)
- Market research (DOCX)

Example Questions:

```
Q: "What were the key financial highlights from Q3?"
Q: "Compare sales performance between different regions"
Q: "What market trends are identified in the research?"
Q: "Based on the data, what are the growth opportunities?"
```

2. Academic Research

Documents to Upload:

- Research papers (PDF)
- Literature reviews (DOCX)
- Data sets (CSV)

Example Questions:

Q: "What are the main research methodologies mentioned?"
Q: "Summarize the key findings across all papers"
Q: "What gaps in research are identified?"
Q: "How do the different studies relate to each other?"

3. Project Management

Documents to Upload:

- Project charters (DOCX)
- Timeline spreadsheets (XLSX)
- Risk assessments (PDF)
- Meeting notes (TXT)

Example Questions:

Q: "What are the critical milestones for this project?"
Q: "List all identified risks and their mitigation strategies"
Q: "What decisions were made in the latest meetings?"
Q: "Are there any conflicting requirements across documents?"

4. Legal Document Review

Documents to Upload:

- Contracts (PDF)
- Legal briefs (DOCX)
- Compliance documents (PDF)

Example Questions:

Q: "What are the key terms and conditions?"
Q: "Identify any potential compliance issues"
Q: "Compare clauses across different contracts"
Q: "What are the termination conditions?"

Advanced Query Techniques

1. Comparative Analysis

"Compare the budget allocations between 2023 and 2024"
"How do the recommendations in document A differ from document B?"
"What are the similarities and differences in the methodologies used?"

2. Trend Analysis

"What trends can you identify in the sales data over time?"
 "How has customer satisfaction changed according to the reports?"
 "What patterns emerge from the research findings?"

3. Synthesis Questions

"Based on all uploaded documents, what is the overall strategy?"
 "Synthesize the key recommendations from all reports"
 "What common themes appear across different documents?"

4. Specific Detail Extraction

"Who are all the stakeholders mentioned in the documents?"
 "List all the action items from the meeting minutes"
 "What specific dates and deadlines are mentioned?"

Optimizing Search Results

Search Depth Settings:

- **Standard (5 sources):** Quick answers, general questions
- **Deep (8 sources):** Detailed analysis, comparative questions
- **Comprehensive (12 sources):** Complex synthesis, research questions

Query Best Practices:

Good Queries:

- "What are the main risks identified in the project plan?"
- "Summarize the financial performance metrics from Q2"
- "Compare the customer feedback themes across regions"

Avoid These:

- "Tell me everything" (too broad)
- "Yes or no" (too simple)
- Single word queries (insufficient context)

Understanding the Interface

Document Upload Tab

- **File Selection:** Supports drag-and-drop and browse
- **Progress Tracking:** Real-time processing status
- **Processing History:** View past uploads and statistics

Query Interface Tab

- **Question Input:** Large text area for detailed questions
- **Search Depth:** Adjustable based on complexity needed
- **Chat History:** Review previous questions and answers

Knowledge Base Tab

- **Statistics Overview:** Document counts and storage info
- **File Listing:** See all uploaded documents
- **Search Function:** Direct content search

Settings Tab

- **Configuration Display:** Current system settings
- **Management Tools:** Clear database, reset application
- **Environment Status:** API key and configuration status

Pro Tips for Power Users

1. Document Preparation

- **Clean Formatting:** Well-formatted documents work better
- **Descriptive Filenames:** Use clear, descriptive file names
- **Reasonable Size:** Keep files under 50MB for optimal performance

2. Query Strategies

- **Start Broad, Then Narrow:** Begin with overview questions
- **Follow-up Questions:** Build on previous answers
- **Context Building:** Reference specific documents when needed

3. Workflow Optimization

- **Batch Upload:** Upload related documents together
- **Progressive Questioning:** Start simple, increase complexity
- **Source Verification:** Always check source attributions

4. Quality Assurance

- **Cross-Reference:** Compare answers with original documents
- **Specificity:** Ask for specific examples or citations
- **Validation:** Use the search function to verify details

Troubleshooting Common Issues

Upload Problems

Issue: "File too large"

Solution: Split large files or reduce file size

Issue: "Unsupported format"

Solution: Convert to supported format (PDF, DOCX, etc.)

Issue: "Processing failed"

Solution: Check file integrity and try again

Query Issues

Issue: "No relevant results"

Solution: Try rephrasing question **or** check **if** documents are uploaded

Issue: "Incomplete answers"

Solution: Increase search depth **or** ask more specific questions

Issue: "Slow responses"

Solution: Check internet connection **and** try simpler queries

Measuring Success

Quality Indicators:

- **Relevance Scores:** Higher scores indicate better matches
- **Source Attribution:** Multiple sources suggest comprehensive answers
- **Answer Completeness:** Detailed responses with examples
- **Consistency:** Similar answers to related questions

Performance Metrics:

- **Response Time:** Should be under 30 seconds for most queries
- **Processing Speed:** Documents should process within minutes
- **Accuracy:** Answers should be verifiable against source documents

Remember: The Agentic RAG Assistant learns from your documents. The more relevant, well-organized content you provide, the better the responses will be!